

Exercise 4: Perform Geospatial Data Visualization using Heatmaps and Choropleth Maps

Aim

To perform geospatial data visualization using Python by creating Heatmaps and Choropleth maps for better business intelligence insights.

Procedure

Install required Python libraries: pandas, matplotlib, folium.

Import necessary libraries in Python.

Load the dataset (CSV file containing location data such as latitude, longitude, and sales).

Check the dataset using head() and info() functions.

Clean the dataset by removing missing or null values.

Create a base map using the Folium library.

Add heatmap layer using latitude and longitude values.

Visualize density using HeatMap plugin.

Create a choropleth map using geojson data.

Map region-wise values (e.g., sales) to color intensity.

Display and save the generated maps as HTML files.

Program (Python)

Copy code

Python

```
# Install if needed:
```

```
# pip install pandas folium matplotlib
```

```
import pandas as pd
```

```
import folium
```

```
from folium.plugins import HeatMap
```

```
# Step 1: Load dataset
```

```
data = pd.read_csv("geospatial_data.csv")
```

Step 2: Remove missing values

```
data = data.dropna()
```

Step 3: Create base map

```
base_map = folium.Map(location=[data['Latitude'].mean(),  
                               data['Longitude'].mean()],  
                      zoom_start=5)
```

Step 4: Create Heatmap

```
heat_data = list(zip(data['Latitude'], data['Longitude'], data['Sales']))  
HeatMap(heat_data).add_to(base_map)
```

Step 5: Save Heatmap

```
base_map.save("heatmap.html")
```

Step 6: Create Choropleth Map

```
choropleth_map = folium.Map(location=[data['Latitude'].mean(),  
                                     data['Longitude'].mean()],  
                             zoom_start=5)
```

```
folium.Choropleth(  
    geo_data="india_states.geojson",  
    data=data,  
    columns=["State", "Sales"],  
    key_on="feature.properties.State",  
    fill_color="YlGnBu",  
    fill_opacity=0.7,  
    line_opacity=0.2,  
).add_to(choropleth_map)
```

```
choropleth_map.save("choropleth_map.html")
```

Output

Result

Geospatial data visualization was successfully performed using Heatmap and Choropleth maps. The visualization helps in identifying high-density business regions and regional performance differences.

Exercise 5: Visualize Data Trends with Bar Charts, Scatter Plots, and Histograms

Aim

To analyze and visualize business data trends using bar charts, scatter plots, and histograms in Python.

Procedure

Import pandas and matplotlib libraries.

Load dataset from CSV file.

Inspect dataset structure.

Handle missing values if present.

Select relevant columns for visualization.

Create bar chart for categorical comparison.

Create scatter plot to observe relationship between two variables.

Create histogram to analyze distribution.

Label axes and titles properly.

Display plots.

Interpret results.

Program (Python)

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Python

```
import pandas as pd

import matplotlib.pyplot as plt


# Load dataset

data = pd.read_csv("sales_data.csv")


# Remove missing values

data = data.dropna()


# Bar Chart

plt.figure()

plt.bar(data['Product'], data['Sales'])

plt.title("Sales by Product")

plt.xlabel("Product")

plt.ylabel("Sales")

plt.xticks(rotation=45)

plt.show()


# Scatter Plot

plt.figure()

plt.scatter(data['Marketing_Spend'], data['Sales'])

plt.title("Marketing Spend vs Sales")

plt.xlabel("Marketing Spend")

plt.ylabel("Sales")

plt.show()
```

```
# Histogram
```

```
plt.figure()
```

```
plt.hist(data['Sales'], bins=10)
```

```
plt.title("Distribution of Sales")
```

```
plt.xlabel("Sales")
```

```
plt.ylabel("Frequency")
```

```
plt.show()
```

Output

The data trends were successfully visualized using different graphical techniques. The charts helped in understanding product performance, correlation between variables, and distribution patterns.

Exercise 6: Explore Datasets using Pandas and Matplotlib

Aim

To explore and analyze datasets using Python libraries such as Pandas and Matplotlib for business intelligence applications.

Procedure

Import required libraries.

Load dataset using `read_csv()`.

Display first few records using `head()`.

Check data types using `dtypes`.

Identify missing values using `isnull().sum()`.

Generate summary statistics using `describe()`.

Perform grouping using `groupby()`.

Sort values using `sort_values()`.

Create basic visualization using `matplotlib`.

Interpret statistical results.

Document findings.

Program (Python)

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Python

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load dataset
```

```
data = pd.read_csv("business_data.csv")
```

```
# Display basic info
```

```
print(data.head())
```

```
print(data.info())
```

```
# Summary statistics
```

```
print(data.describe())
```

```
# Group by category
```

```
grouped = data.groupby('Category')['Revenue'].sum()
```

```
print(grouped)
```

```
# Plot grouped data
```

```
plt.figure()
```

```
grouped.plot(kind='bar')
```

```
plt.title("Total Revenue by Category")
```

```
plt.xlabel("Category")
```

```
plt.ylabel("Revenue")
```

```
plt.show()
```

Output

Result

The dataset was successfully explored using Pandas functions and visualized using Matplotlib. Key statistical insights and trends were identified, supporting business intelligence decision-making.

Sample data set;

OrderID	Date	Region	SalesPerson	Product	Units	UnitPrice	Discount
---------	------	--------	-------------	---------	-------	-----------	----------

1001	2024-01-03	North	Alice	Widget	10	9.99	0
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1002	2024-01-05	South	Bob	Gadget	5	19.95	5
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1003	2024-02-10	East	Charlie	Widget	2	9.99	0
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1004	2024-02-20	West	Denise	Thingamajig	12	4.5	2
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1005	2024-03-01	North	Alice	Gadget	1	19.95	0
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1006	2024-03-15	South	Bob	Widget	7	9.99	0
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